

TIM 80C, Lecture #14 (5/14/2026)

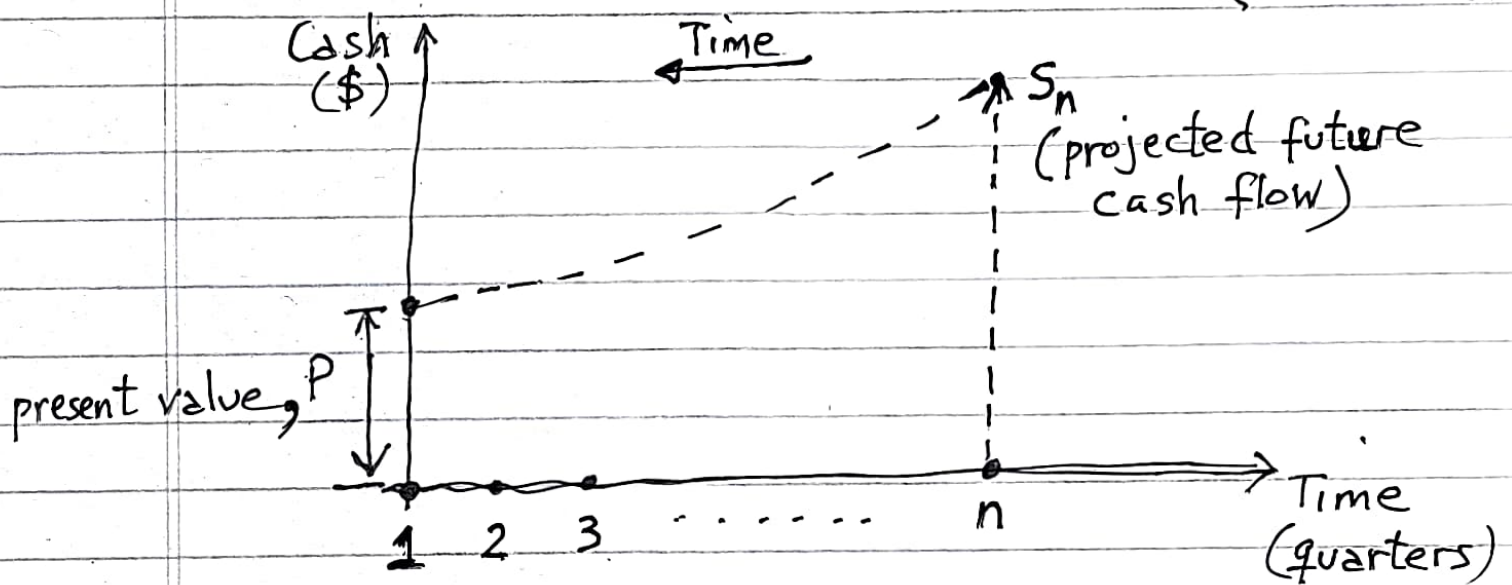
Agenda:

1. Review the Present Value (PV) & the Net Present Value (NPV) equations
2. Another example of determining the expected profit for new product development in a start-up.
3. Financial structure of a company
4. Financial Accounting for a start-up
5. HW #4 has been posted & you should be actively working on it!

1. Review of Present Value (PV) & Net Present Value equations

Annual discount factor = $d\%$ ($= 10\%$, say)

Quarterly discount factor = $\frac{d}{4}\%$ ($= 2.5\%$)
($\equiv 0.025$)



S_n = future cash flow at period (quarter) n

Quarter 1 $\equiv$ Present

P = present value of S_n

$$P = \frac{S_n}{\underbrace{\left(1 + \frac{d}{4}\right) \left(1 + \frac{d}{4}\right) \cdots \left(1 + \frac{d}{4}\right)}_{(n-1) \text{ terms}}} = \frac{S_n}{\left(1 + \frac{d}{4}\right)^{n-1}}$$

Net Present Value (NPV) = Sum of all the present values for each quarter

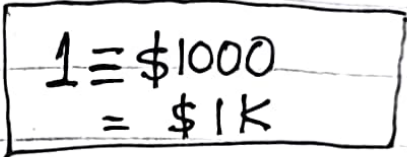
$$= \sum_{i=1}^n (PV)_{Qtr i}$$

2. Another example of computing Expected Profit
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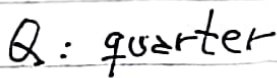
Problem : What is the expected value of profit for new product development in a start-up

Plan (or Process or Approach) & Implementation
for constructing a Base Case or Nominal
model.

time



($\Rightarrow$ simple "add / subtract")



Step 3 : Compute the present value (PV) of the net cash flow (or profit) for each qtr.

Assume an annual discount factor, $d = 10\%$

$$\text{Quarterly discount factor} = \frac{d}{4} = \frac{10}{4} = 2.5\% = 0.025$$

Quarter	Q1	Q2	Q3	Q4	Q5
Net cash flow (from step 2)	-5,000	-5,000	3,000	4,000	5,000
	the "present"				
Present value of each cash flow	-5,000	$\frac{-5,000}{(1+\frac{d}{4})^1}$	$\frac{+3,000}{(1+\frac{d}{4})^2}$	$\frac{+4,000}{(1+\frac{d}{4})^3}$	$\frac{+5,000}{(1+\frac{d}{4})^4}$
		= -4,878	= +2,855	= +3,714	= +4,529

Step 4 : Compute the Net Present Value (NPV) of the expected profit by adding (up) all the individual present values

From the table

$$\begin{aligned} \text{NPV} &= -5,000 + (-4,878) + 2,855 + 3,714 + 4,529 \\ &= \$1,221. \equiv \$1,221 \text{ K} \equiv \$1,221,000 \\ &= \$1.22 \text{ M} \end{aligned}$$

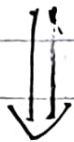
Result : The net present value (NPV) of the expected profit for the start-up based on a time-horizon of 5 quarters, & annual discount factor of 10%, is \$1.22M $\Rightarrow$ good!

For a more complete example, refer to the "Economic / Financial Analysis" handout on Canvas $\Rightarrow$ Polaroid Company, discussed in L#13

2. Financial Structure of a company

2.1 Financing : Companies (aka firms) are financed from 2 broad sources

The firm's owners
(aka share holders)



Cash contributed
by the share holders
(or owners) is called
the EQUITY capital

those who lend
money to the firm
(aka Debt holders)



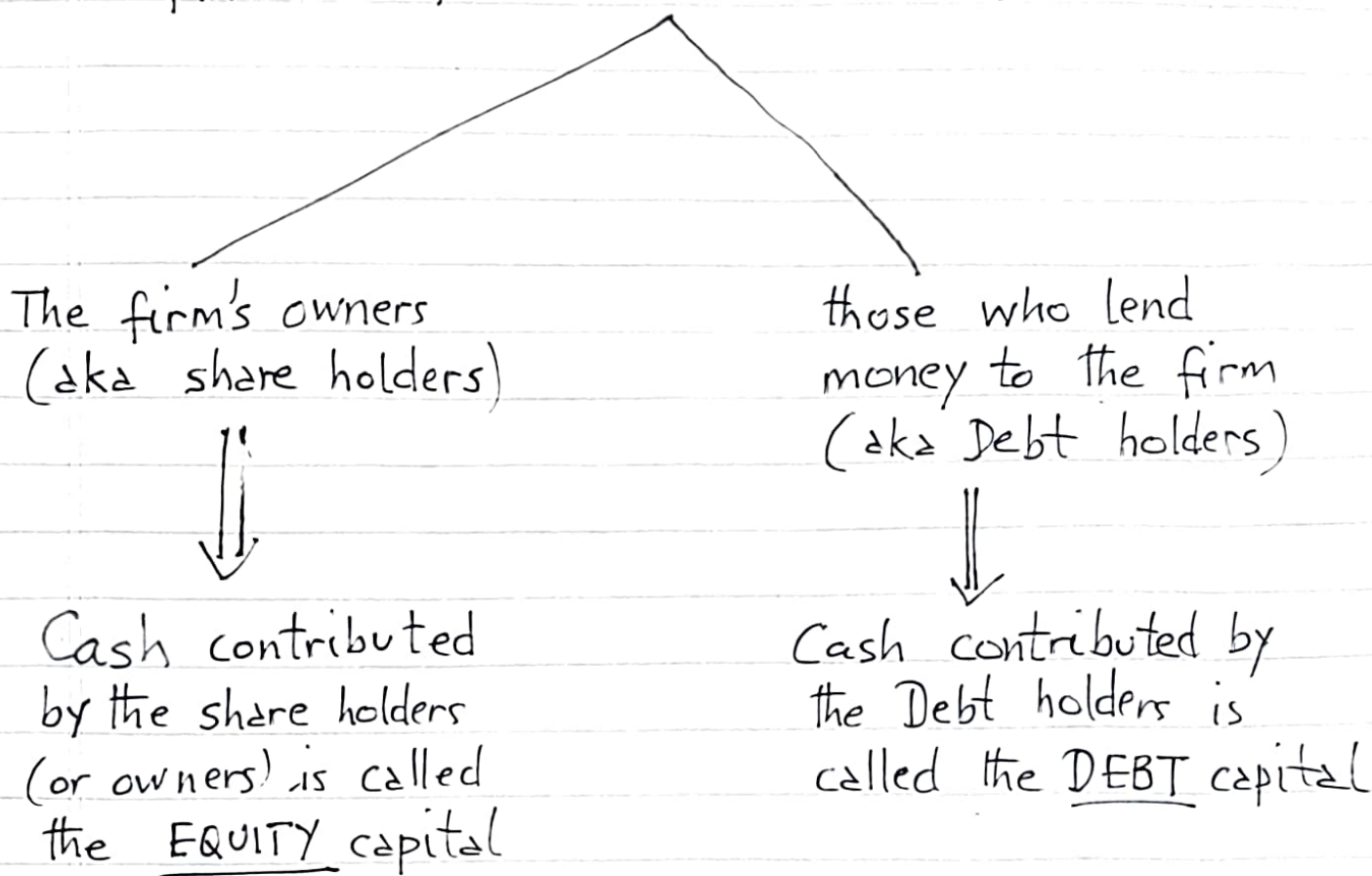
Cash contributed by
the Debt holders is
called the DEBT capital

(i) The capital structure of the firm is
the ratio of the Debt capital
Equity capital

(ii) Asset or total capital = Equity capital + D

2. Financial Structure of a company

2.1 Financing : Companies (aka firms) are financed from 2 broad sources



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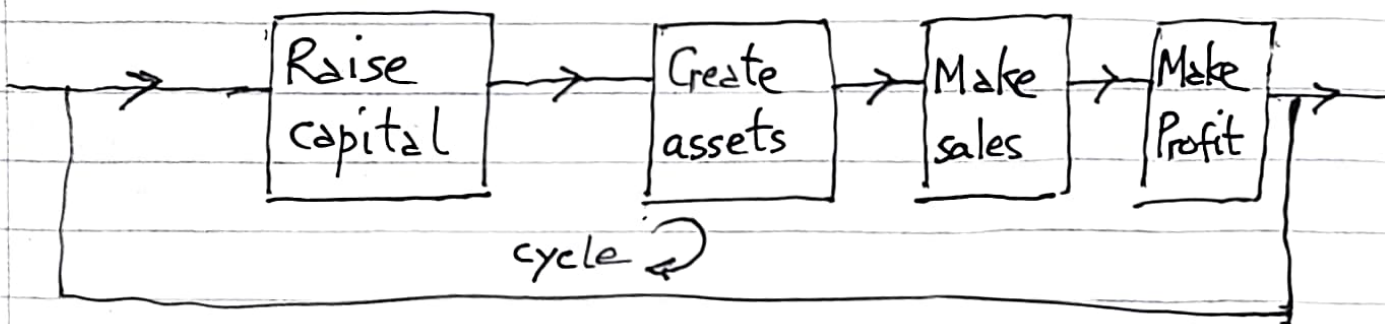
(ii) Asset or total capital = Equity capital + Debt capital

2.2 The fundamental financial principle

A business proposal (e.g., starting a new company) will increase the firm's value if the Net Present Value (NPV) of all actual and expected cash flows is greater than zero.

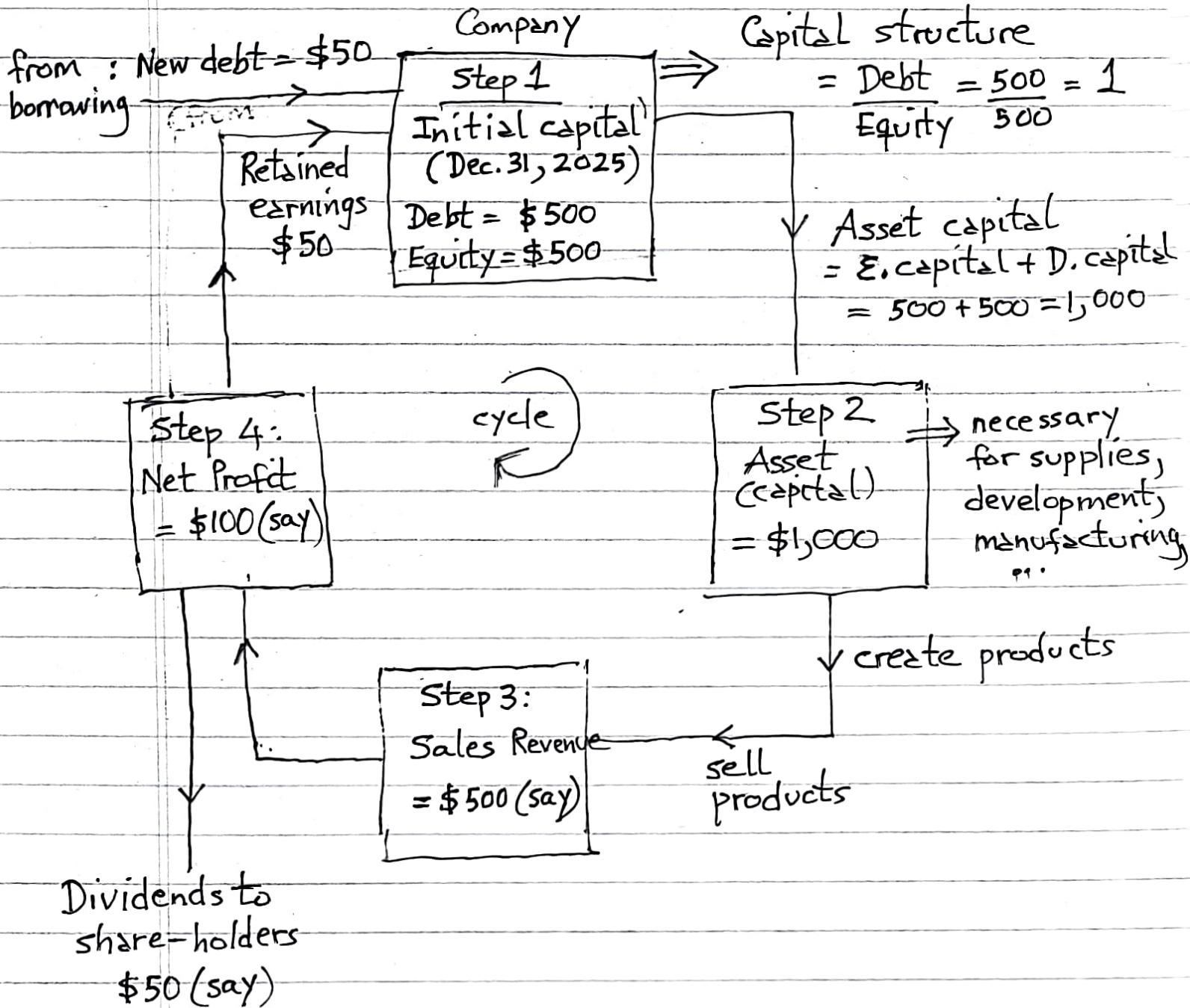
$NPV > 0 \Rightarrow$ proposal adds value
 $NPV < 0 \Rightarrow$ proposal reduces value

2.3 Business Cycle:



Assets : cash, People, infrastructure, producing products ("goods"),

Example (all numbers are in \$1000s)



Important financial ratios

$$\textcircled{1} \text{ Capital structure} = \frac{\text{Debt}}{\text{Equity}} = \frac{\$500}{\$500} = 1.0$$

$\Rightarrow$ Smaller the better!

$$\textcircled{2} \text{ Efficiency of asset usage} = \frac{\text{Sales Revenue}}{\text{Asset capital}}$$

$$= \frac{\$500}{\$1000}$$

$$= 0.5 \text{ or } 50\% \\ (\text{excellent!})$$

$$\textcircled{3} \text{ Profit Margin (\%)} = \frac{\text{Net Profit}}{\text{Sales Revenue}} \times 100\% \\ (\text{aka Return on Sales})$$

$$= \frac{\$100}{\$500} \times 100\% = 20\% \\ (\text{excellent})$$

Net Profit $\equiv$ Net Income

Return $\equiv$ Profit

$$\textcircled{4} \text{ Retention Rate (\%)} = \frac{\text{Retained Earnings}}{\text{Net Profit}} \times 100\%$$

$$= \frac{\$50}{\$100} \times 100\% = 50\%$$

⑤ Return on Invested capital (ROIC) %

$$= \frac{\text{Net Profit}}{\text{Invested capital} (\equiv \text{asset capital})} \times 100\%$$

$$= \frac{\$100}{\$1000} \times 100\% = 10\% \quad \underbrace{\text{v. good.}}$$

⑥ Return on Equity (ROE) %

$$= \frac{\text{Net Profit}}{\text{Shareholder's Equity}} \times 100\%$$

$$= \frac{\$100}{\$500} \times 100\% = 20\% \quad \underbrace{\text{v. good}}$$

⑦ Economic Value Added (EVA) % $\equiv \text{ROIC} - \underbrace{\text{WACC}}_{\substack{\text{weighted} \\ \text{average} \\ \text{cost of capital}}}$

$\equiv$ average cost of borrowing money

WACC is the average % cost of borrowing money

In our case

$$ROIC = 10\% \text{ (see above)}$$

if $WACC = 9\% \text{ (say)}$

then $EVA(\%) = ROIC - WACC = 10 - 9 = 1\%$

$\Rightarrow$ firm is adding value

Principle of the FIRM :

For a company to be successful,
 $EVA(\%)$ should be greater than zero.
Otherwise, economic value is being destroyed.

$EVA > 0 \Rightarrow$ firm is adding value

$EVA < 0 \Rightarrow$ firm is destroying value

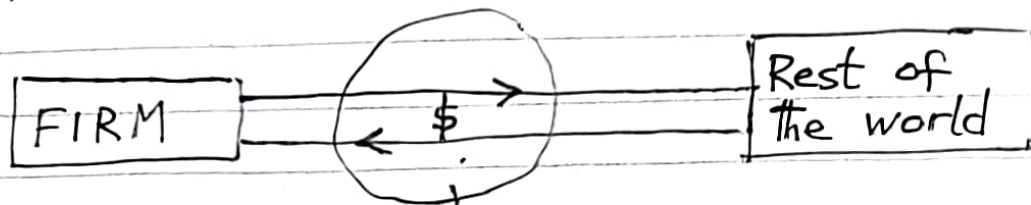
TIM 80C, Lecture #15 (5/19/2026)

Agenda

- 0 : complete "Financial Ratios" development
(done; see above)
- 1 : Financial Accounting
- 2 : Return graded midterms
- 3 : Read the "Adobe founder" (Geschke)
interview on Canvas before Thursday's
class meeting;
share at least one lesson (you)
learned in-class on Thursday
+ 1 pt. (on HW#4) for sharing
- 1 pt. (on HW#4) for not sharing
(HW # 4, Prob. 5)
4. Midterm re-submission due
Thursday, this week, 5/21/2026
5. HW # 4 is now due on Tuesday, 5/26/26

1. Financial Accounting

A simplified view of financial accounting for this course



Financial Transactions



Financial Accounting Process

Balance sheet:
Record of the firm's assets and liabilities at the date of the balance sheet

Income statement:
Record of all the revenues & expenses, over a period of time.

Cash Flow statement:
statement of all the cash flows over a period of time

1.1 Balance sheet

The balance sheet is a statement that shows, at a specific point in time (say, Dec. 31, 2025), the following:

- (i) Assets : cash, inventory of products, facilities, equipment,
- (ii) Liabilities : money that the firm owes to the bank, suppliers,

The owner's equity is the difference between Assets & Liabilities

$$\text{Equity} = \text{Assets} - \text{Liabilities}$$

$$\underbrace{\text{Assets} = \text{Liabilities} + \text{Equity}}$$

"Balance"

Example : A very simple balance sheet
for the ACME Company: 31 Dec. 2025
(all numbers are in \$1000s)

Assets :	Liabilities :
Cash : \$ 500	short term loans : \$ 200 (e.g., to pay suppliers)
Fixed assets : \$ 500 (equipment, inventory, ...)	long term loans : \$ 500 (e.g., mortgages, ...)
① Total assets = \$1,000	② Total Liabilities : \$ 700
	③ Owner's Equity : \$ 300 (shareholder's equity)

Assets = Liabilities + Owner's Equity

"Balance"